Average Rate of Change

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.03

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Computing average rate of change

Find the average rate of change of $f(x) = x^2$ over each interval:

a $[0, 2]$ **b** $[1, 3]$ **c** $[2, 5]$ **d** $[-1, 1]$

2 [Foundational] • Target: Computing average rate of change

Find the average rate of change of $f(x) = 2^x$ over the interval $[0, 3]$.

3 [Foundational] • Target: Computing average rate of change

Show that the average rate of change of $g(x) = 3x - 5$ is the same over any interval $[a, b]$, and state its value.

4 [Foundational] • Target: Applications

A ball is thrown upward so that its height after t seconds is $h(t) = -5t^2 + 20t$ meters.

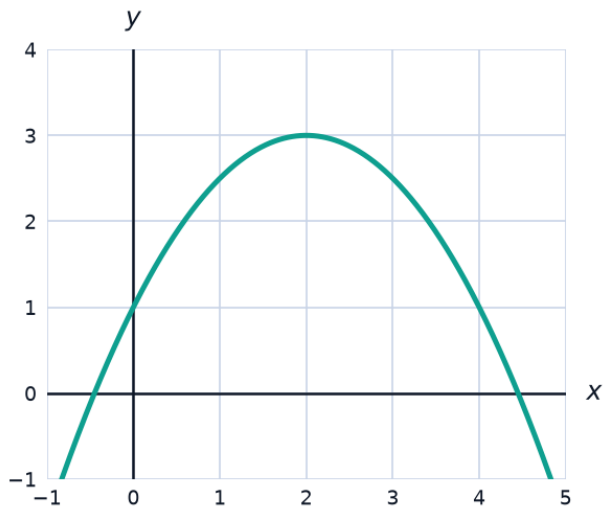
a Find the average rate of change of the height over $[0, 2]$.

b Find the average rate of change of the height over $[1, 3]$.

c Interpret your answer to part b in the context of the ball's motion.

5 [Foundational] • Target: Applications

The graph of $f(x) = -\frac{1}{2}x^2 + 2x + 1$ is shown. Estimate the average rate of change over $[0, 4]$, then verify your estimate algebraically.



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6 [Foundational] • Target: Applications

A company's revenue (in thousands of riyals) is modeled by $R(t) = 40 + 6t - 0.5t^2$, where t is years since 2020.

a Find the average rate of change of revenue over $[0, 4]$.

b Find the average rate of change of revenue over $[4, 8]$.

c Compare your two answers and interpret what they say about the company's growth.

7 [Foundational] • Target: Comparing rates and estimating from a table

A table of values for $g(x)$ is given: $g(1) = 5$, $g(3) = 17$, $g(6) = 44$, $g(10) = 92$. Find the average rate of change of g over:

a $[1, 3]$ **b** $[3, 6]$ **c** $[6, 10]$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Comparing rates and estimating from a table

Based on the previous question, is the average rate of change of g increasing, decreasing, or constant as x increases? What does this suggest about the shape of g's graph?

9 [Intermediate] • Target: Comparing rates and estimating from a table

Two functions model distance traveled: $d_1(t) = 5t$ and $d_2(t) = t^2$ (in km, t in hours). Find the average speed (average rate of change) of each over $[2, 6]$, and state which traveler was faster on average.

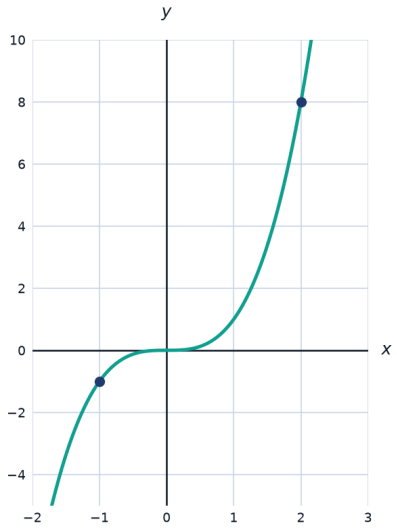
10 [Intermediate] • Target: Average rate of change and secant lines

Explain the geometric meaning of the average rate of change of f over $[a, b]$ in terms of the secant line through $(a, f(a))$ and $(b, f(b))$.

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11 **[Intermediate]** • **Target: Average rate of change and secant lines**

For $f(x) = x^3$, find the average rate of change over $[-1, 2]$, and write the equation of the secant line through the corresponding points.



12 **[Intermediate]** • **Target: Comparing average rate of change across different functions**

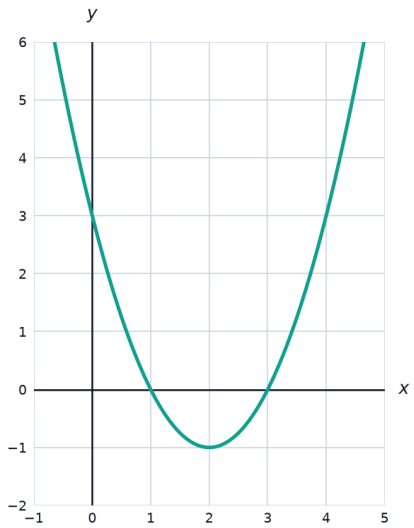
Find the average rate of change of $f(x) = 2^x$ and $g(x) = x^2$ over $[0, 4]$, and determine which function grew faster on average.

13 **[Intermediate]** • **Target: Comparing average rate of change across different functions**

Find the average rate of change of $f(x) = 2^x$ over $[4, 8]$ and compare it to its value over $[0, 4]$ (found above). What does this reveal about exponential growth?

14 **[Intermediate]** • **Target: Average rate of change from a graph**

Using the graph of $f(x) = x^2 - 4x + 3$ shown, estimate the average rate of change over $[0, 4]$ by reading endpoint values from the graph, then confirm algebraically.



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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Units and interpretation in context • Cross-Lesson: 6.01 Polar Coordinates and Conversions

A tank's water volume is $V(t) = 200 - 3t^2$ liters, where t is minutes. Find the average rate of change over $[0, 5]$, including correct units, and interpret its sign.

16 [Challenging] • Target: Units and interpretation in context • Cross-Lesson: 6.02 Polar Equations and Graphs

A population model gives $P(t) = 500e^{0.1t}$ (people, years). Find the average rate of change over $[0, 10]$ and state its units, rounding to the nearest whole number.

17 [Challenging] • Target: Comparing rates and estimating from a table • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

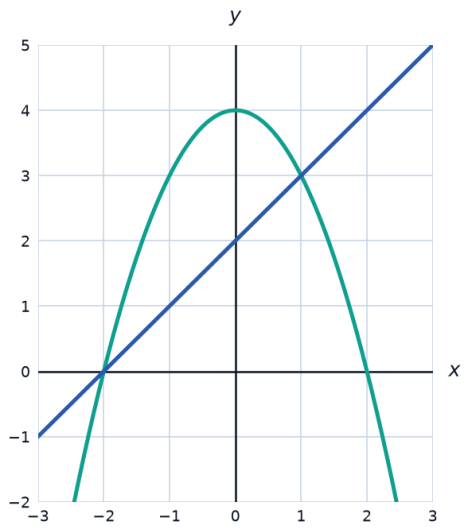
Given vectors $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = -3\mathbf{i} + 4\mathbf{j}$, find the scalar projection of $\mathbf{a}$ onto $\mathbf{b}$ and the angle θ between them to the nearest tenth of a degree.

18 [Challenging] • Target: Average rate of change and secant lines • Cross-Lesson: 7.02 Logarithms and Their Properties

Eliminate the parameter to determine the Cartesian equation of the curve $x = 1 + 2\sec t$, $y = -2 + 3\tan t$. State the shape, center, and vertices.

19 [Challenging] • Target: Comparing average rate of change across different functions • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

Find the average rate of change of the polynomial function $f(x) = 2x^3 - 5x + 4$ over the closed interval $[-1, 2]$.



20 [Challenging] • Target: Average rate of change from a graph • Cross-Lesson: 7.04 Applications: Growth and Decay

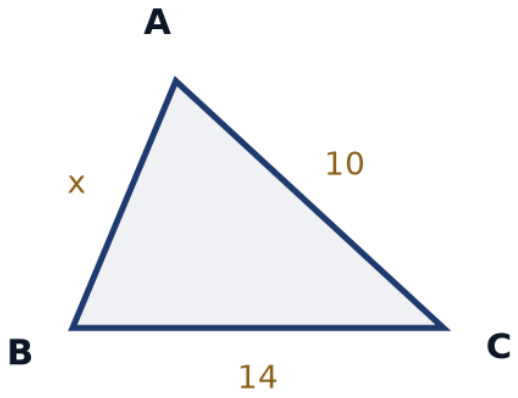
[Vectors, Lines & Parametric Kinematics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Computing average rate of change

In triangle ABC, angle A is 52 degrees. The side opposite angle B (AC) has a length of 10 units, and the side opposite angle C (AB) has a length of 14 units. What is the length of the side opposite angle A (BC)?



a. 10.3

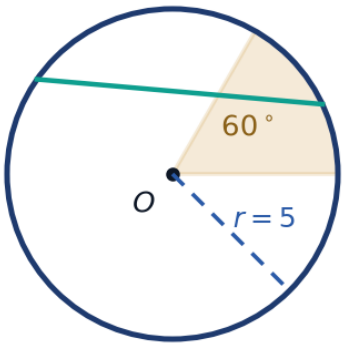
b. 12.1

c. 14.8

d. 8.5

22 [Foundational] • Target: Applications

A circle has a radius of 6 units. A chord within this circle subtends a sector with a central angle of 60 degrees. What is the length of the chord?



a. 6 units

b. $6\sqrt{3}$ units

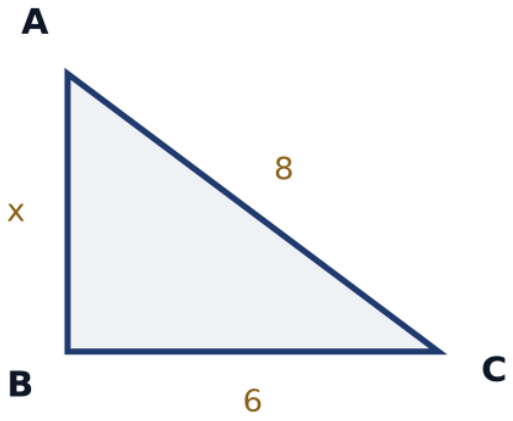
c. 12 units

d. $12\sqrt{3}$ units

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23 [Foundational] • Target: Comparing rates and estimating from a table

In a right-angled triangle ABC, the angle B is 90 degrees. If the length of side AB is 8 units and the length of side BC is 6 units, what is the length of the hypotenuse AC?



a. 10 units

b. 14 units

c. 48 units

d. 20 units

24 [Foundational] • Target: Average rate of change and secant lines

$f(x) = 2x^2 + 2x - 5$. Find the average rate of change of f on $[1, 3]$.

a. 10

b. -10

c. 9

d. 5

25 [Foundational] • Target: Comparing average rate of change across different functions

Two particles, A and B, move in the xy-plane. Their position vectors at time t seconds, for $t \geq 0$, are given by $\mathbf{r}_A(t) = (2t; 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t; 1 - 2t)$ respectively. The particles start their motion at $t = 0$. Find the minimum distance between particle A and particle B.

a. $\sqrt{26}$

b. $2\sqrt{13}$

c. $3\sqrt{10}$

d. $4\sqrt{5}$

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26 [Foundational] • Target: Average rate of change from a graph

Consider a circle with its center at the origin $(0, 0)$ and a radius of 5 units. A chord of this circle is perpendicular to the x-axis and intersects the x-axis at the point $(3, 0)$. The endpoints of this chord on the circle are A and B. Determine the length of the chord AB.

a. 6

b. 8

c. 10

d. 4

27 [Foundational] • Target: Units and interpretation in context

The graph of the function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of the parameter a.

a. 1

b. 2

c. 3

d. 4

28 [Foundational] • Target: Computing average rate of change

Consider the exponential growth model $P(t) = P_0 e^{kt}$, where $P(t)$ is the population at time t and P_0 is the initial population. If the initial population is $P_0 = 500$ and after 10 years the population is $P(10) = 1200$, find the annual growth rate k. Round your answer to four decimal places.

a. 0.0588

b. 0.0875

c. 0.1200

d. 0.0434

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Applications

Consider two vectors, $\mathbf{u} = (3; k)$ and $\mathbf{v} = (-2; 4)$. If $\mathbf{u}$ is parallel to $\mathbf{v}$, find the value of the scalar parameter k.

a. -6

b. 6

c. 3

d. -3

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30 [Intermediate] • Target: Comparing rates and estimating from a table

In triangle ABC, side $a = 10$ units and side $b = 15$ units. If the angle $C = 60^\circ$, find the length of side c. Round your answer to two decimal places.

a. 13.23

b. 14.70

c. 12.96

d. 15.52

31 [Intermediate] • Target: Average rate of change and secant lines

Consider two points in the Cartesian plane, $P = (2, 3)$ and $Q = (5, 7)$. What is the component form of the vector $\vec{PQ}$?

a. $\langle 3, 4 \rangle$

b. $\langle -3, -4 \rangle$

c. $\langle 7, 10 \rangle$

d. $\langle 1, 4 \rangle$

32 [Intermediate] • Target: Comparing average rate of change across different functions

Two vectors are given by $\vec{u} = 3\hat{i} + k\hat{j}$ and $\vec{v} = -\hat{i} + 4\hat{j}$. If $\vec{u}$ and $\vec{v}$ are orthogonal, find the value of the scalar parameter k.

a. 3/4

b. 4/3

c. -3/4

d. -4/3

33 [Intermediate] • Target: Average rate of change from a graph

Consider the vector $\mathbf{v} = \langle 3, 4 \rangle$. What is the magnitude of this vector?

a. 5

b. 7

c. 12

d. 25

34 [Intermediate] • Target: Units and interpretation in context

Consider the vector $\mathbf{v}$ with initial point $P = (2, 3)$ and terminal point $Q = (5, 7)$. What are the components of the vector $\mathbf{v}$?

a. ■3, 4■

b. ■7, 10■

c. ■-3, -4■

d. ■4, 3■

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35 [Intermediate] • Target: Computing average rate of change

Find the period and phase shift of $y = \tan(1/2x - 1)$.

a. Period 2π, phase shift 0

b. Period 2π, phase shift 2 units left

c. Period π, phase shift 2 units right

d. Period 2π, phase shift 2 units right

36 [Intermediate] • Target: Applications

A population of 500 grows at 5% per year, modeled by $P(t) = 500(1 + 0.05)^t$. Approximately how many years until the population reaches $1.5\times$ its initial size (i.e. $P = 1.5 \cdot 500$)?

a. 8.3 years

b. 10.8 years

c. 6.3 years

d. 10.3 years

37 [Intermediate] • Target: Comparing rates and estimating from a table

Given $f(x) = (x - 2)^2(x + 1)$, does the graph touch or cross the x-axis at $x = 2$?

a. The graph does not intersect the x-axis there

b. Not enough information

c. Touches (does not cross)

d. Crosses

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Average rate of change and secant lines • Cross-Lesson: 7.02 Logarithms and Their Properties

Find the reference angle for 220°.

a. 60°

b. 40°

c. 50°

d. 140°

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39 [Challenging] • Target: Comparing average rate of change across different functions • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

What are the maximum and minimum values of $y = 3\sin(x)$?

a. Max 2, Min -2

b. Max 3, Min -3

c. Max 0, Min 0

d. Max 4, Min -4

40 [Challenging] • Target: Average rate of change from a graph • Cross-Lesson: 7.04 Applications: Growth and Decay

What is the amplitude and period of $y = 5\sin(3x) - 3$?

a. Amplitude -3, period $\frac{2}{3}\pi$

b. Amplitude 5, period $\frac{2}{3}\pi$

c. Amplitude 5, period $\frac{3}{2}\pi$

d. Amplitude 5, period 2π

41 [Challenging] • Target: Units and interpretation in context • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Write $5^4 = 625$ in logarithmic form.

a. $\log_{625}(5) = 4$

b. $\log_5(4) = 625$

c. $\log_4(5) = 625$

d. $\log_5(625) = 4$

42 [Challenging] • Target: Computing average rate of change • Cross-Lesson: 8.02 Graphing Rational Functions

Convert 150° to radians.

a. $\frac{5\pi}{6}$

b. $\frac{2\pi}{3}$

c. $\frac{\pi}{6}$

d. $\frac{7\pi}{4}$

43 [Challenging] • Target: Applications • Cross-Lesson: 8.03 Introduction to Limits

A principal of 1000 earns compound interest at rate 10% per period for 2 period(s), compounded annually. Find the final amount.

a. 1215

b. 1200

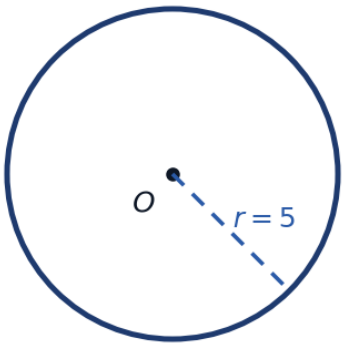
c. 1210

d. 1000

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44 [Challenging] • Target: Comparing rates and estimating from a table • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.



a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$

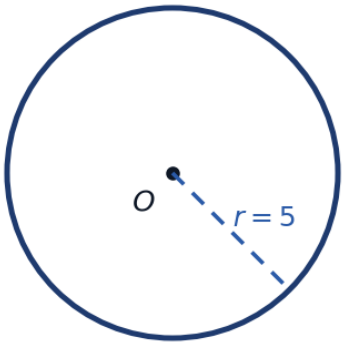
c. 1

d. $\frac{\sqrt{2}}{2}$

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45 [Challenging] • Target: Average rate of change and secant lines • Cross-Lesson: 9.02 Geometric Sequences and Series

A circle is centered at the origin and passes through the point $P(a, 4)$ in the first quadrant. The equation of a tangent line to the circle at point P is given by $3x + 4y = k$. Find the value of k.



a. 25

b. 30

c. 35

d. 40

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